

# **Cross-Asset and Sector-Wide Analysis to Predict Suncor Returns**

Group 3: Andrea Abenoja (30021197) and Bhuvan Dubey (30139765)

**DATA 611 Predictive Analysis**

Project 2

Semester: Winter 2026

Date: February 25, 2026

## INTRODUCTION

As one of Canada's highest revenue industries, the oil and gas sector is a key contributor to the country's energy and economic output. As of 2026, gasoline and petroleum bulk stations, wholesaling, oil drilling and gas extraction, and petroleum refining comprises the top 10 industries by revenue in Canada (Government of Canada, 2024; IBISWorld, n.d.). Suncor, an energy company founded in Canada, has operations across all of these sectors and continues to play a major role in the country's energy landscape. In November 2025, Suncor released their third-quarter financial results during which Suncor CEO, Rich Kruger, announced that he "strongly believes in buying our stock is our best investment" (Suncor Energy, 2025), signaling confidence in the company's financial outlook and drawing attention to the value of accurately predicting Suncor's stock performance.

Despite this internal optimism, stock returns are inherently influenced by broader market conditions and even more so in commodity-driven industries. Fluctuations in crude oil prices directly affect upstream financial performance, while movements in the USD/CAD exchange rate influence both revenue translation and the international competitiveness of Canadian producers. In addition, peer companies often exhibit correlated return behavior due to shared exposure to commodity cycles, regulatory environments, and capital investment trends. To capture these sector-wide dynamics, this study incorporates return signals from Shell, Imperial Oil, Cenovus Energy, and Canadian Natural Resources Limited (CNRL). Cenovus, Imperial Oil, and Canadian Natural Resources are consistently ranked among the largest oil and gas producers in Canada by market capitalization and production capacity, making them particularly relevant benchmarks for domestic sector performance, while Shell provides an international integrated energy comparator (Carpenter, n.d.).

By modeling Suncor's daily returns against crude oil prices, exchange rate shifts, and the performance of its major peers, the objective of this project aims to utilize supervised learning methods, multiple linear regression and random forest regression to test whether these broader market signals can improve the prediction of Suncor daily returns. Not only does this analysis help provide more insight into industry trends, it can provide incremental predictive value and enhance forecast accuracy for investors and analysts.

## METHODS

### Reference Dataset

Daily closing prices for all variables were retrieved using the GOOGLFINANCE function in Google Sheets, covering the period from January 2, 2015 to January 1, 2026. The tickers used were TSE:SU (Suncor), NYSEARCA:USO (United States Oil Fund), CURRENCY:USDCAD (USD to CAD Foreign Exchange), NYSE:SHEL (Shell), TSE:IMO (Imperial Oil), TSE:CVE (Cenovus), and TSE:CNQ (CNRL). As Google Finance does not support direct retrieval of WTI crude oil prices, the United States Oil Fund (USO), an exchange-traded fund tracking oil futures, was used as a proxy. A standardized date table spanning the full period was constructed and XLOOKUP was used to align closing prices across all variables, as the raw data retrieved varied in length across tickers. Daily returns were then calculated as the percentage change in closing price from the previous trading day, using the formula:

$$\text{Daily Return} = \frac{\text{Current Day's Closing Price} - \text{Previous Day's Closing Price}}{\text{Previous Day's Closing Price}}$$

Non-trading days, which appeared as blank values, were assigned a daily return of zero, reflecting the assumption of no price change when markets are closed. The final dataset retained only daily return columns and was exported with formula values preserved for use in Python. Additionally, the resulting data table had 8 columns: Date, suncor\_daily\_return, USOF\_daily\_return, USDtoCAD\_daily\_return, shell\_daily\_return, imperial\_oil\_daily\_return, cenovus\_daily\_return, cnrl\_daily\_return, and 2,769 observations corresponding to the daily returns from January 2, 2015 to January 1, 2026.

### Analyses

Two predictive models were developed to estimate Suncor's daily returns: a multiple linear regression (MLR) and a random forest model. MLR was selected as a baseline interpretable model to examine the linear relationships between Suncor's returns and its predictors. Random forest was selected as a non-linear alternative capable of capturing more complex relationships among predictors that MLR may not detect, while also providing feature importance rankings to identify which predictors contribute most to the model. The random forest classification model was trained using 100 estimators, with the square root of the total number of features considered at each split (`max_features = 'sqrt'`) and a minimum of 5

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observations required at each leaf node ( $\text{min\_samples\_leaf} = 5$ ). These hyperparameters were set to balance model complexity and prevent overfitting. A random state of 42 was set to ensure reproducibility. For both models, the dataset was split chronologically, with 70% of observations used for training and the remaining 30% reserved for testing, preserving the time series structure of the data and preventing data leakage.

Model performance was evaluated using RMSE and adjusted  $R^2$  on the test set, where RMSE measures average prediction error and adjusted  $R^2$  reflects the proportion of variance explained while accounting for the number of predictors. In addition to MLR, a binary classification model was developed using random forest to predict the direction of Suncor's daily return (e.g. whether the return would be positive or negative). The classification model was evaluated using the Area Under the ROC Curve (AUC), where a value closer to 1 indicates stronger ability to correctly distinguish between positive and negative return days.

## RESULTS

### Multiple Linear Regression Analysis

A multiple linear regression model was used to predict Suncor daily returns using 6 predictors: USOF daily returns, USD/CAD daily returns, and the daily returns of Shell, Imperial Oil, Cenovus, and CNRL. The model was trained on 70% of the observations and evaluated on the remaining 30%. The model had an adjusted  $R^2$  of 0.755 on the test data, suggesting that the model performed well on test data as the model was able to explain 75.5% of the variation in Suncor's daily returns. As well, the model's prediction error ( $\text{RMSE}=0.00630$ ) was 50.6% lower than the baseline prediction error ( $\text{RSME}=0.01280$ ). Overall, the model was statistically significant ( $p < 0.05$ ) and 71.1% of the variation in the training data was explained by the model ( $\text{adjusted } R^2=0.711$ ).

Out of the six predictors, only 5 were statistically significant at  $p<0.05$ : USD/CAD, Imperial Oil, Shell, Cenovus and CNRL. From these 5 predictors, Imperial Oil had the largest coefficient ( $\beta = 0.3309$ ), followed by CNRL ( $\beta = 0.2854$ ), Shell ( $\beta = 0.1749$ ), Cenovus ( $\beta = 0.1414$ ) and USD/CAD ( $\beta = 0.2854$ ). USOF was the only predictor that was not statistically significant ( $p=0.667$ ). Multicollinearity was not observed in the predictors as all variance inflation factors (VIF) were between 1.00 and 3.00. A correlation matrix (Figure 1c) also showed no multicollinearity between predictors. However, the model's errors were not normally distributed as visualized in the Q-Q Plot (Figure 1a). The Q-Q Plot of the residuals show a

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substantial deviation away from the normal lines at both tail ends. As well, the residuals vs predicted values plot (Figure 1b) did not show a funnel shape indicative of heteroscedasticity, residuals were randomly scattered around zero across the predicted values. The Cook's Distance analysis (Figure 1d) identified 5 observations with influence on the model.

Figure 1 illustrates the fit of the predicted values, in orange, in comparison to the actual values, in blue. The model was generally accurate at predicting the day to day movement of Suncor's returns. However, the model consistently underestimated during periods of extreme market movement, such as big losses or big gains.

### Random Forest Regression Analysis

Figure 3 shows how well the Random Forest regression model predicted the next day returns for Suncor. This graph shows the actual returns (blue line) and the predicted returns (orange line) over the test period. The two lines are almost identical, which means that the model is a good fit for the data. The Root Mean Squared Error (RMSE) is 0.006265 which is very small. A small RMSE shows how well the predictions are to the actual returns, and that the model has strong predictive capabilities for Suncor's next-day returns.

The feature importance scores obtained from the Random Forest model are shown in Figure 4a. It shows the most important factors influencing the daily returns of Suncor. Imperial Oil's daily return shows the highest importance score at 0.3185, indicating that it is the most significant predictor among all variables. CNRL's daily return also plays a very large role, with an importance score of 0.2508, followed by Cenovus's daily return at 0.2214. Shell's daily return has a lower importance score of 0.1236 but remains a notable factor. External market signals contribute less to the model's predictions, with the US Oil Fund's daily return at 0.0624 and the USD/CAD exchange rate's daily return at 0.0233, reflecting their comparatively minor influence. Complementing this, Figure 4b illustrates the partial dependence plots for these top features, showing how changes in each factor affect Suncor's returns. The plots reveal that as Imperial Oil's daily return increases, Suncor's return tends to rise correspondingly, with a similar pattern observed for CNRL and Cenovus. Shell's partial dependence indicates a positive relationship, though less pronounced. The US Oil Fund and USD/CAD exchange rate show weaker, more gradual effects, aligning with their lower importance scores. Together, these figures demonstrate that sector peers, especially Imperial Oil, CNRL, and Cenovus, are the primary drivers

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influencing Suncor's daily returns, while external market indicators have a comparatively smaller impact.

The ROC curve in Figure 5 illustrates the binary classification task of identifying whether Suncor's return is positive or negative on the next trading day using the same set of external variables. The Area Under the Curve (AUC) measure of 0.964 in the ROC curve indicates that the model has a good discriminative power for distinguishing between positive and negative returns. The high AUC value also indicates that the model has the ability to accurately predict the direction of Suncor's next-day return, which is extremely useful for trading purposes. The steep rise of the ROC curve towards the top-left corner also indicates the model's excellent predictive capability.

## DISCUSSION

### Multiple Linear Regression Analysis

The multiple linear regression model demonstrated strong predictive performance, explaining 75.5% of the variation in Suncor's daily returns, cutting prediction error in half compared to baseline. These results indicate that peer Canadian oil company returns and the USD to CAD exchange rate are meaningful drivers to Suncor's daily returns. This is likely explained by the fact that these companies operate in the same industry and are affected by the same oil prices and economic conditions.

Within-sector predictors appeared to have the strongest association to Suncor's returns, as Imperial Oil had the largest coefficient, followed by CNRL, Shell and Cenovus. These companies share key characteristics with Suncor (e.g. oil sands operations, Canadian-based revenues, and similar cost structures) which can help to explain their strong co-movement. USD to CAD exchange rate was also a significant predictor as Suncor earns revenue in USD, but has the majority of their costs in CAD, making returns sensitive to currency fluctuations. USOF was the only insignificant predictor, which may be due to the fact that oil prices may already be captured in peer stock returns.

Most of the model's assumptions were satisfied as multicollinearity was not observed and there was no evidence of heteroscedasticity. However, the model violated the normality assumption as the residuals deviated from the theoretical normality line in the QQ Plot seen in Figure 1a. This indicated that the errors are more extreme than expected in a normal distribution,

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which is a common behaviour seen in financial data (Fama, 1965). While the model was able to predict day to day movement relatively well, the model struggled during periods of extreme Suncor-specific movement as demonstrated by the underestimation of returns in Figure 2 and the influential points observed in the Cook's Distance plot in Figure 1d.

#### Random Forest Regression Analysis

This study provides valuable insights into the factors influencing Suncor's daily returns and the predictive capabilities of machine learning models in financial markets. The results indicate that sector peers, particularly Imperial Oil, CNRL, and Cenovus, are the most significant predictors of Suncor's next-day returns. External market signals such as the US Oil Fund and USD/CAD exchange rate, while included in the models, have comparatively minor impacts. The Partial Dependence plots further clarify that increases in the returns of key sector companies are associated with corresponding rises in Suncor's returns, emphasizing the interconnectedness within the sector. The high AUC score of 0.964 demonstrates that the classification model is highly effective in predicting the direction of Suncor's next-day returns, offering practical utility for trading strategies.

The results also emphasize the importance of nonlinear models such as Random Forests in understanding the complex relationships between external signals and peer stock movements that influence the returns of Suncor. The importance of peer stocks indicates the influence of sector-wide factors, and the high ROC AUC indicates the potential of the model for directional forecasting, which helps traders and investors in making informed decisions. The results indicate the importance of cross-asset and sector-wide signals in containing predictive information, particularly when modeled using nonlinear techniques.

However, there are some challenges and limitations that the study encounters. The first challenge is that the study is based on historical data, which may not be able to capture the unexpected shocks in the market. The models that are used are robust, but they have limitations in terms of the assumptions that are made and the quality of the data that is used. The external factors that are difficult to quantify and may not have been captured in the models include geopolitical events, policy changes, and supply disruptions. The study is limited to a specific set of external variables and a specific time period, which may not be generalizable to other market conditions and time periods.

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**CONCLUSION**

The multiple linear regression model proved to be an effective approach to predicting Suncor's daily returns, with within-sector peer companies and the USD to CAD exchange rate emerging as the primary drivers. The model was able to explain 75.5% of the variation in Suncor's returns on test data and reduced prediction error by over 50% compared to baseline. That said, the violation of the normality assumption and the presence of influential observations emphasize the model's limitations in capturing extreme Suncor-specific market movements. Future studies could consider alternative modelling approaches that explicitly account for volatility and extreme market events can be used to address these limitations.

This project also shows the great potential of machine learning models, especially nonlinear models such as Random Forests, in stock return prediction and the identification of the complex relationships between external market indicators and sector peers. The results emphasize the major role of sector-related stocks like Imperial Oil, CNRL, and Cenovus in Suncor's performance, as well as the important but less significant role of external factors. The high accuracy and discriminative ability indicate that combining such models with trading strategies could improve decision-making and risk management. The limitations in terms of data coverage and market unpredictability indicate the need for further improvement and the incorporation of more macroeconomic variables. Machine learning models provide promising directions for financial forecasting but must be used in an integrated and dynamic strategy to deal with unpredictable markets.



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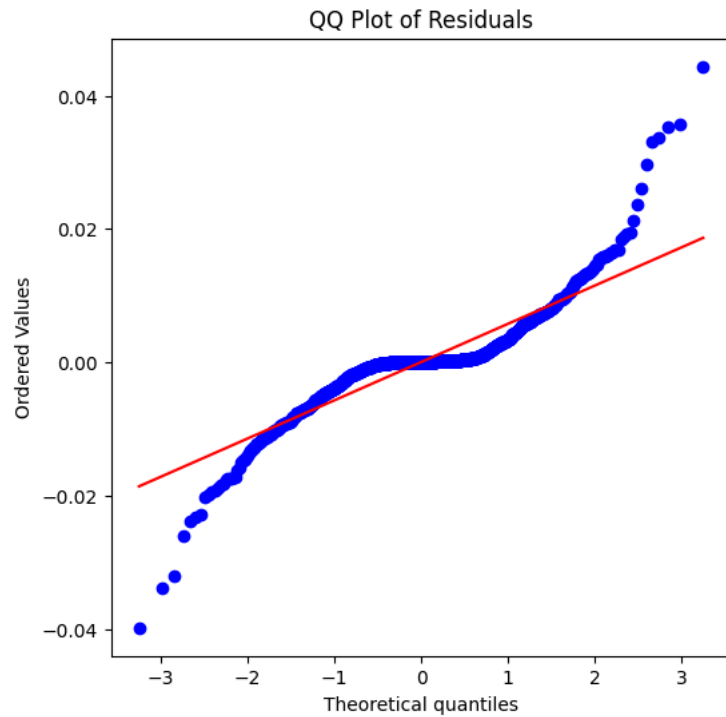
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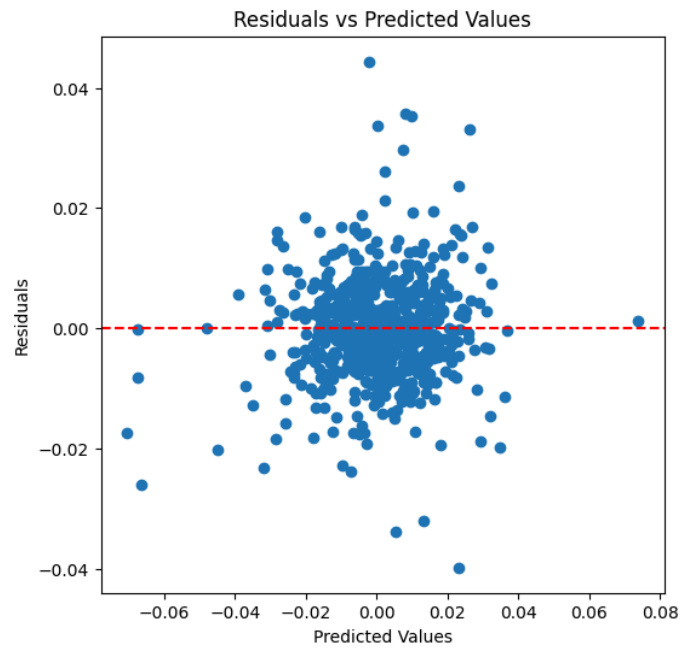
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### Appendix



**Figure 1a.** Q-Q Plot of Residuals to Check for Normality of Residuals Assumption



**Figure 1b.** Residuals vs Predicted Values to Check for Heteroscedasticity Assumption

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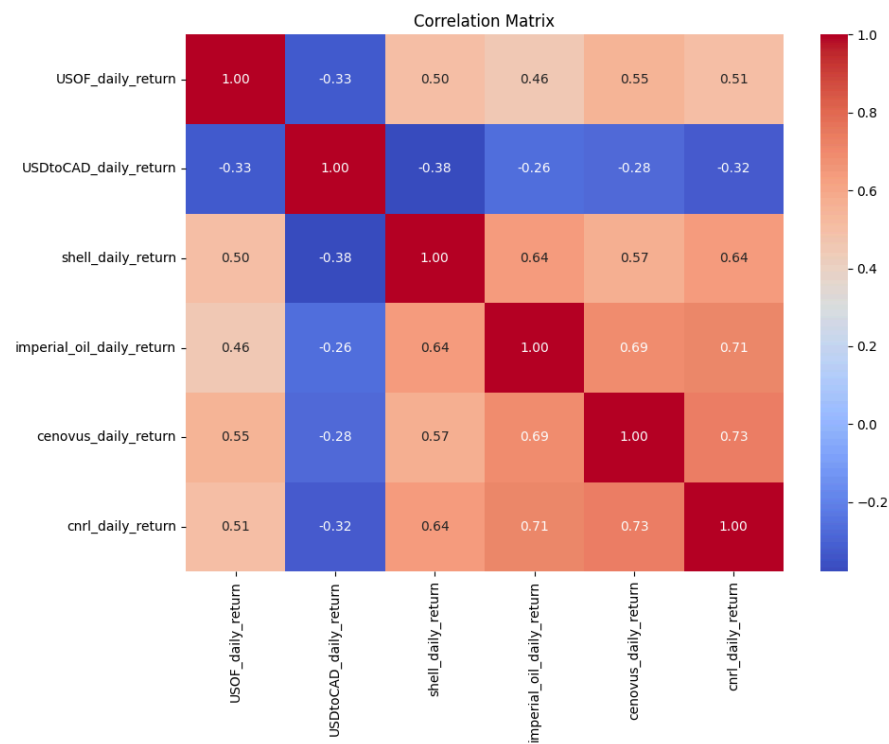


Figure 1c. Correlation Matrix to Check for Multicollinearity Assumption

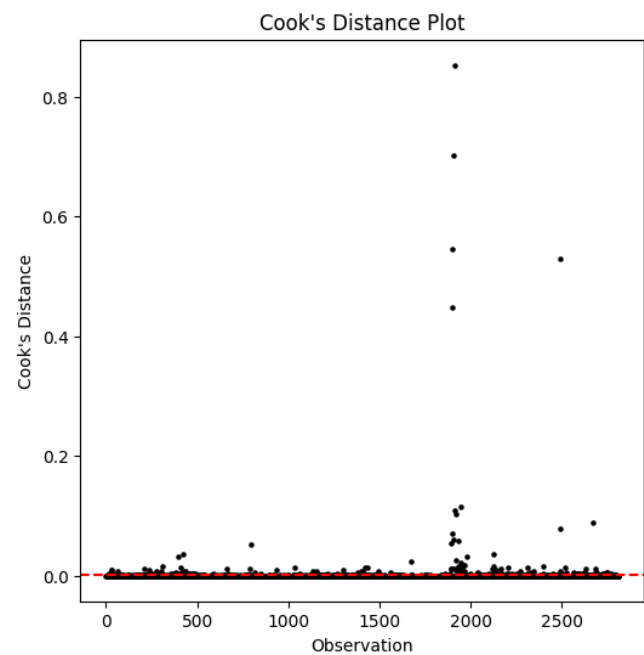
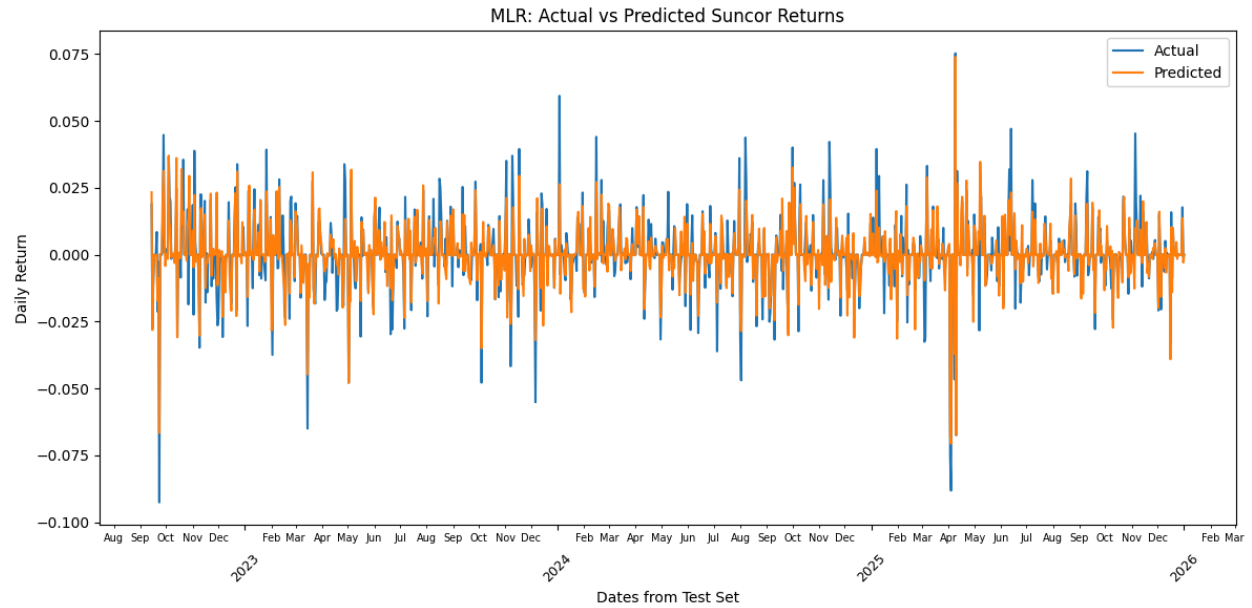
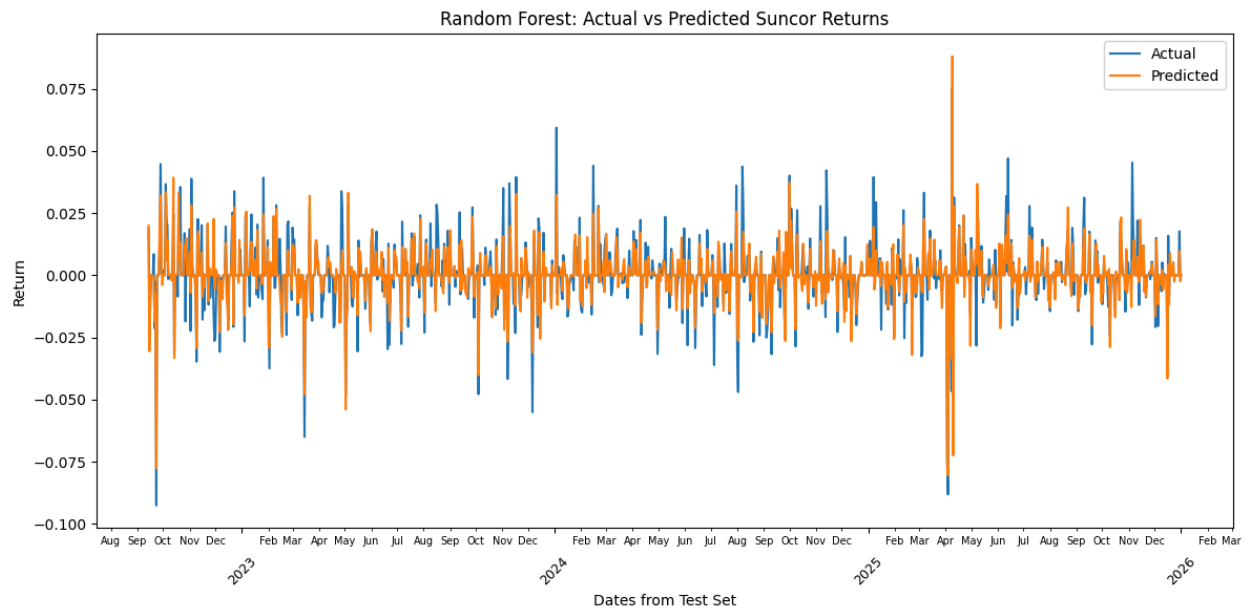


Figure 1d. Cook's Distance Plot to Check for Influential Points Assumption

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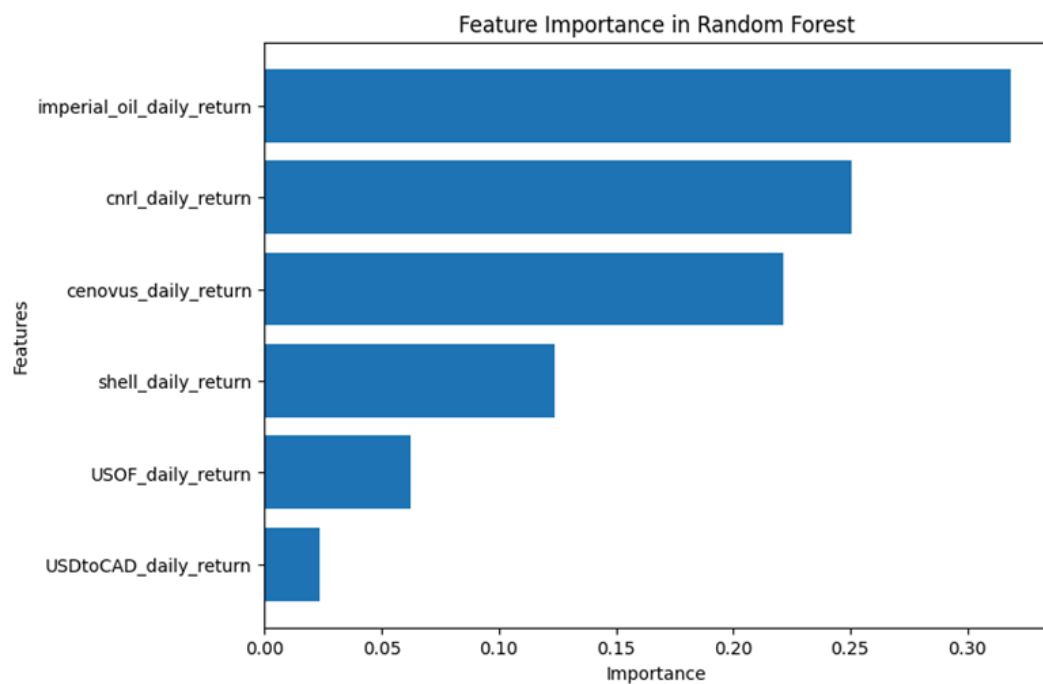
**Figure 2.** Actual vs. Predicted Suncor Returns Using Multiple Linear Regression



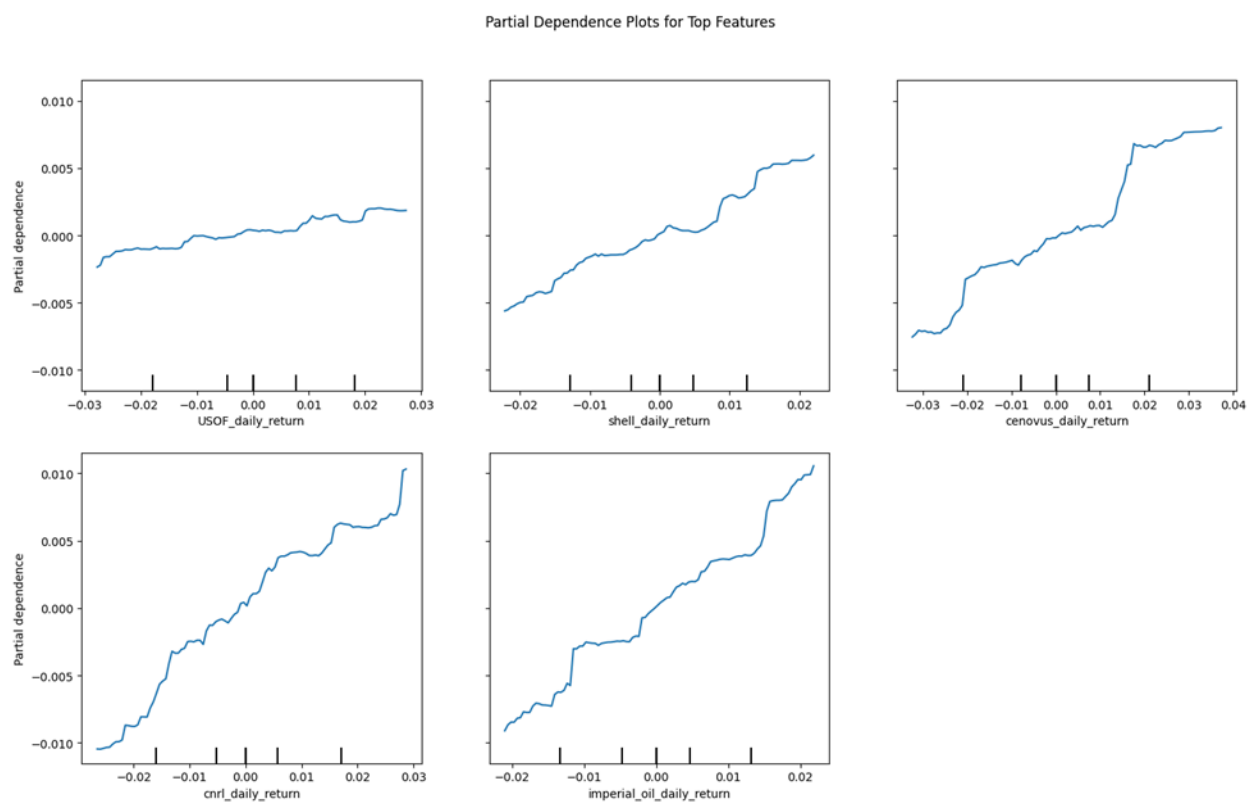
**Figure 3.** Actual vs. Predicted Suncor Returns Using Random Forest Regression

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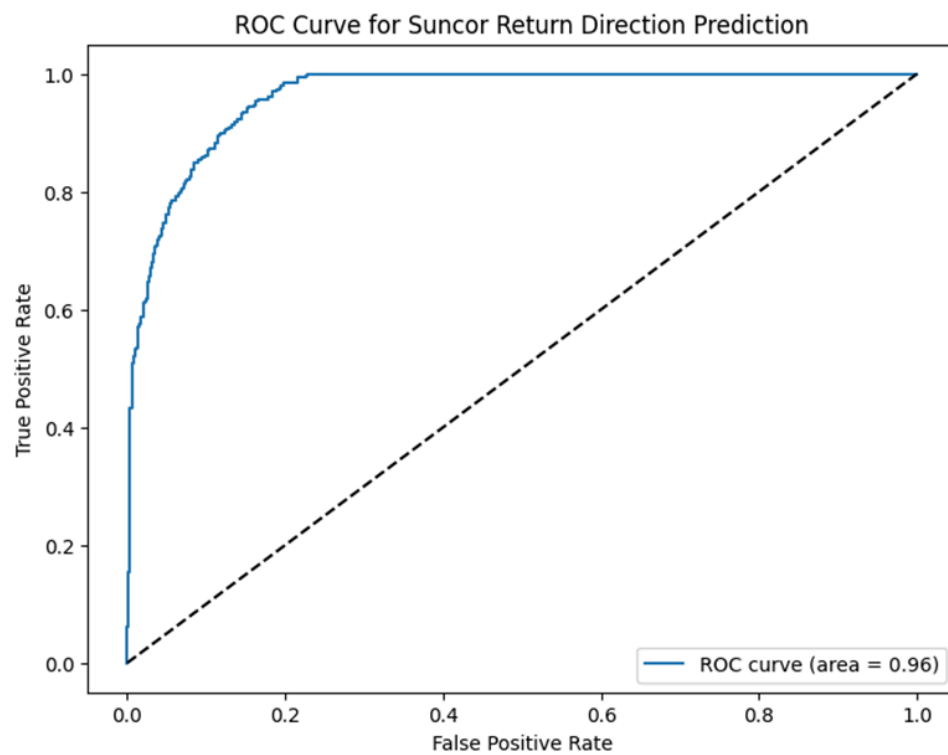
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**Figure 4a.** Feature Importance in the Random Forest Model



**Figure 4b.** Partial Dependence Plots for Top Features



**Figure 5.** ROC Curve for Suncor Return Direction Prediction